

Project Report

On

SKYHIGH AIRLINES PERFORMANCE ANALYTICS USING POWER BI

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TABLE OF CONTENTS

Abstract	iii
1. Introduction	iii
2. Problem Statement	iii
3. Objectives	iii
4. Tools and Technologies	iv
4.1 Software Used.....	iv
4.2 Data Sources.....	iv
5. Data Modeling	iv
5.1 Relationship Created	iv
5.2 Benefits.....	iv
6. DAX Calculations	v
7. Results and Analysis	vi
7.1 Dashboard 1: Executive Summary.....	vi
7.2 Dashboard 2: Flight Operations.....	vii
7.3 Dashboard 3: Customer Insights.....	viii
8. Advantages	ix
9. Future Enhancements	ix
10. Conclusion	ix
11. References	ix

ABSTRACT

The airline industry generates a significant volume of operational, financial, and customer-related data. Analyzing this data effectively helps improve operational efficiency, passenger satisfaction, and profitability. This project presents an interactive Power BI dashboard for SkyHigh Airlines to analyze flight performance, delays, revenue trends, airport operations, and passenger satisfaction. The dashboard provides management with meaningful insights through Cards, visual reports, and business intelligence techniques to support data-driven decision-making.

Keywords:

Power BI, Business Intelligence, Data Visualization, DAX, Airline Analytics, Dashboard.

1 INTRODUCTION

Business Intelligence (BI) plays a vital role in transforming raw data into meaningful insights. Airlines manage complex operations involving flight scheduling, airport management, customer service, and revenue optimization. Monitoring these operations manually is time-consuming and inefficient. This project utilizes Power BI to build an Airline Performance Analytics Dashboard that consolidates flight, passenger, airport, and revenue data into a single reporting platform. The solution provides real-time insights into airline operations and customer satisfaction, enabling management to make informed strategic decisions.

2 PROBLEM STATEMENT

SkyHigh Airlines aims to improve operational efficiency and customer satisfaction by analyzing airline data. Existing reporting methods do not provide sufficient visibility into flight performance, delays, passenger feedback, and revenue trends. Therefore, a centralized Power BI dashboard is required for effective monitoring and decision-making.

3 OBJECTIVES

1. Analyze overall airline performance.
2. Track flight punctuality and delay trends.
3. Evaluate passenger satisfaction scores.
4. Identify top-performing routes and airports.
5. Analyze revenue growth patterns.
6. Support management decision-making using Cards and dashboards.

4 TOOLS AND TECHNOLOGIES

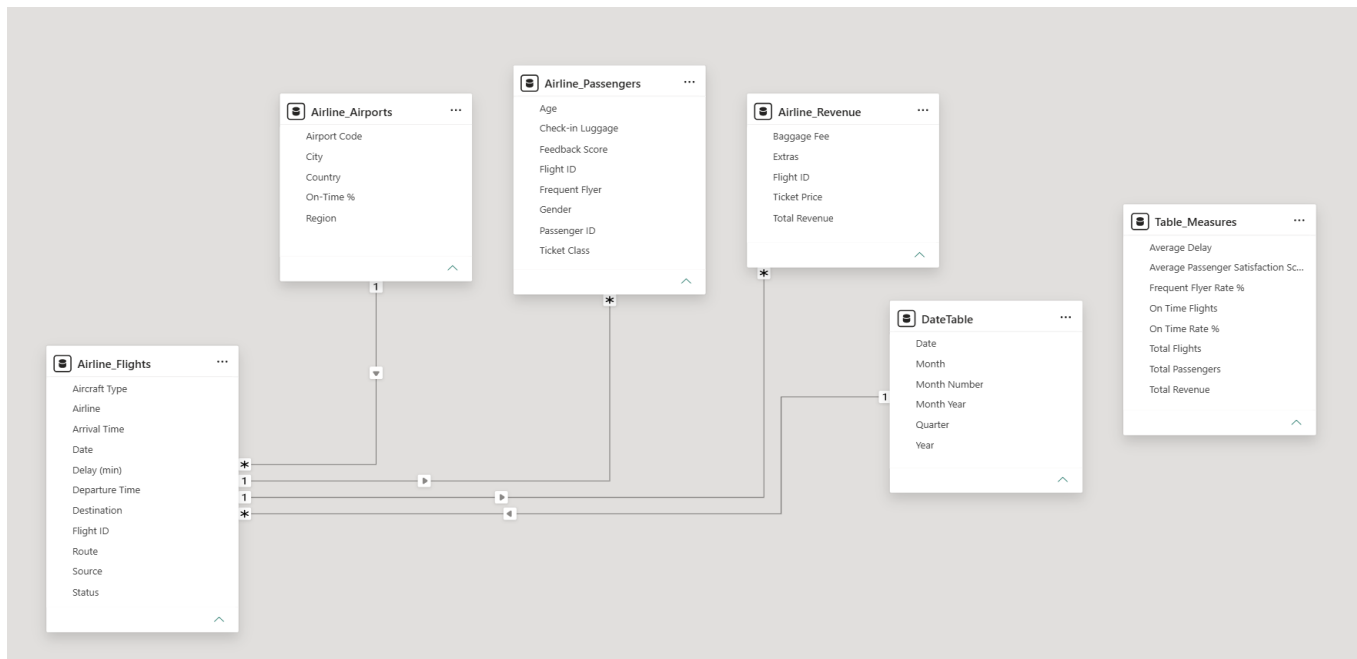
4.1 Software Used:

- Microsoft Power BI Desktop
- Power Query Editor
- DAX (Data Analysis Expressions)

4.2 Data Sources:

- Airline_Flights.csv
- Airline_Passengers.csv
- Airline_Revenue.csv
- Airline_Airports.csv

5 DATA MODELING



5.1 Relationships Created:

- Flights (Flight ID) → Passengers (Flight ID)
- Flights (Flight ID) → Revenue (Flight ID)
- Flights (Source) → Airports (Airport Code)
- Flights (Date) → DateTable (Date)

5.2 Benefits:

- Improved query performance.
- Consistent reporting.
- Better data integrity.

6 DAX CALCULATIONS

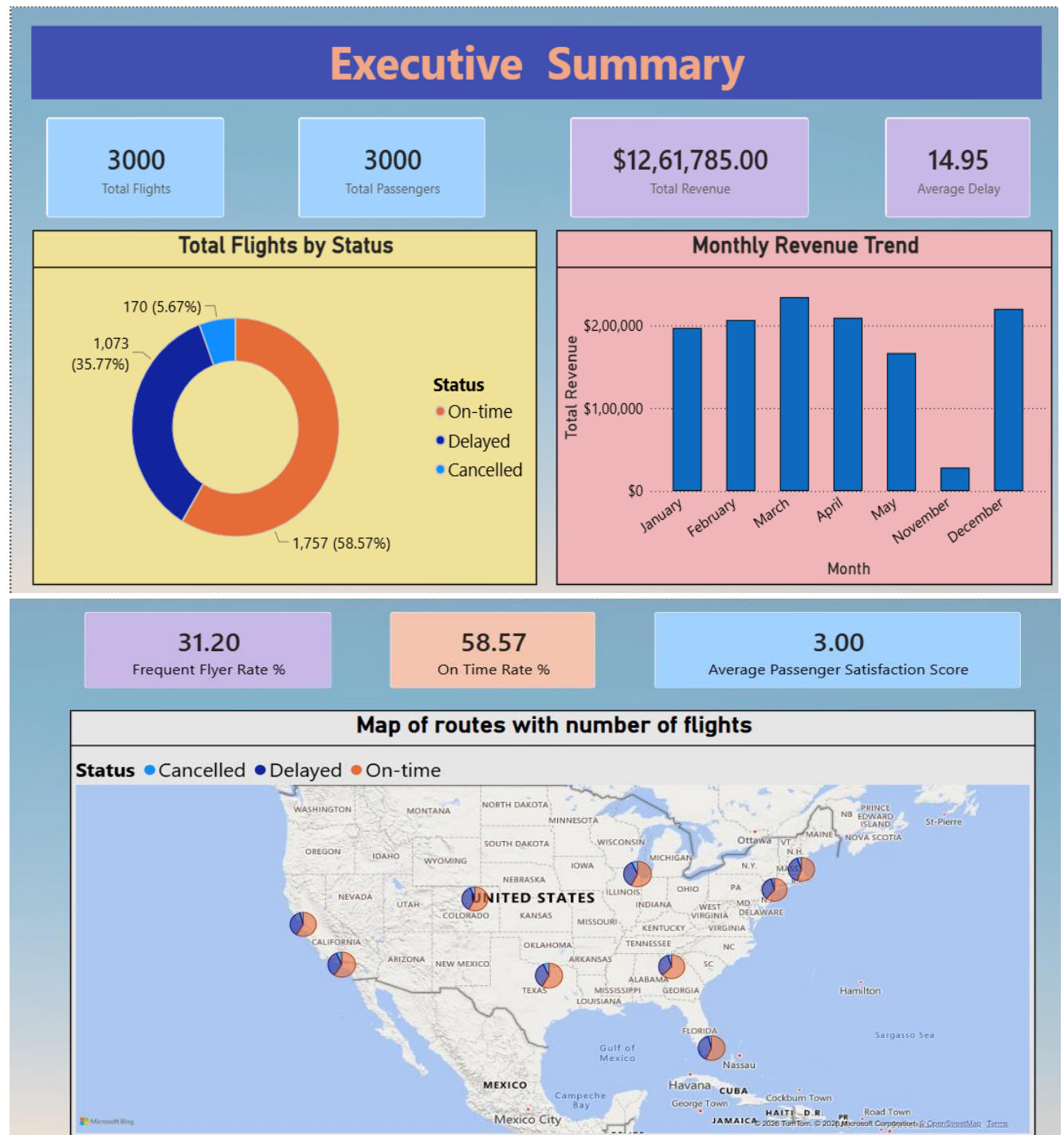
- I. Total Flights = **COUNT**(Airline_Flights[Flight ID])
- II. Total Passengers = **COUNT**(Airline_Passengers[Passenger ID])
- III. Total Revenue = **SUM**(Airline_Revenue[Total Revenue])
- IV. Average Delay = **AVERAGE**(Airline_Flights[Delay (min)])
- V. Frequent Flyer Rate % = **DIVIDE**(**CALCULATE**(**COUNTROWS**(Airline_Passengers), Airline_Passengers[Frequent Flyer] = "Y"), **COUNTROWS**(Airline_Passengers)) * 100
- VI. On Time Rate % = **DIVIDE**([On Time Flights],[Total Flights]) * 100
- VII. Average Passenger Satisfaction Score = **AVERAGE**(Airline_Passengers[Feedback Score])



7 RESULTS AND ANALYSIS

- Dashboard Development

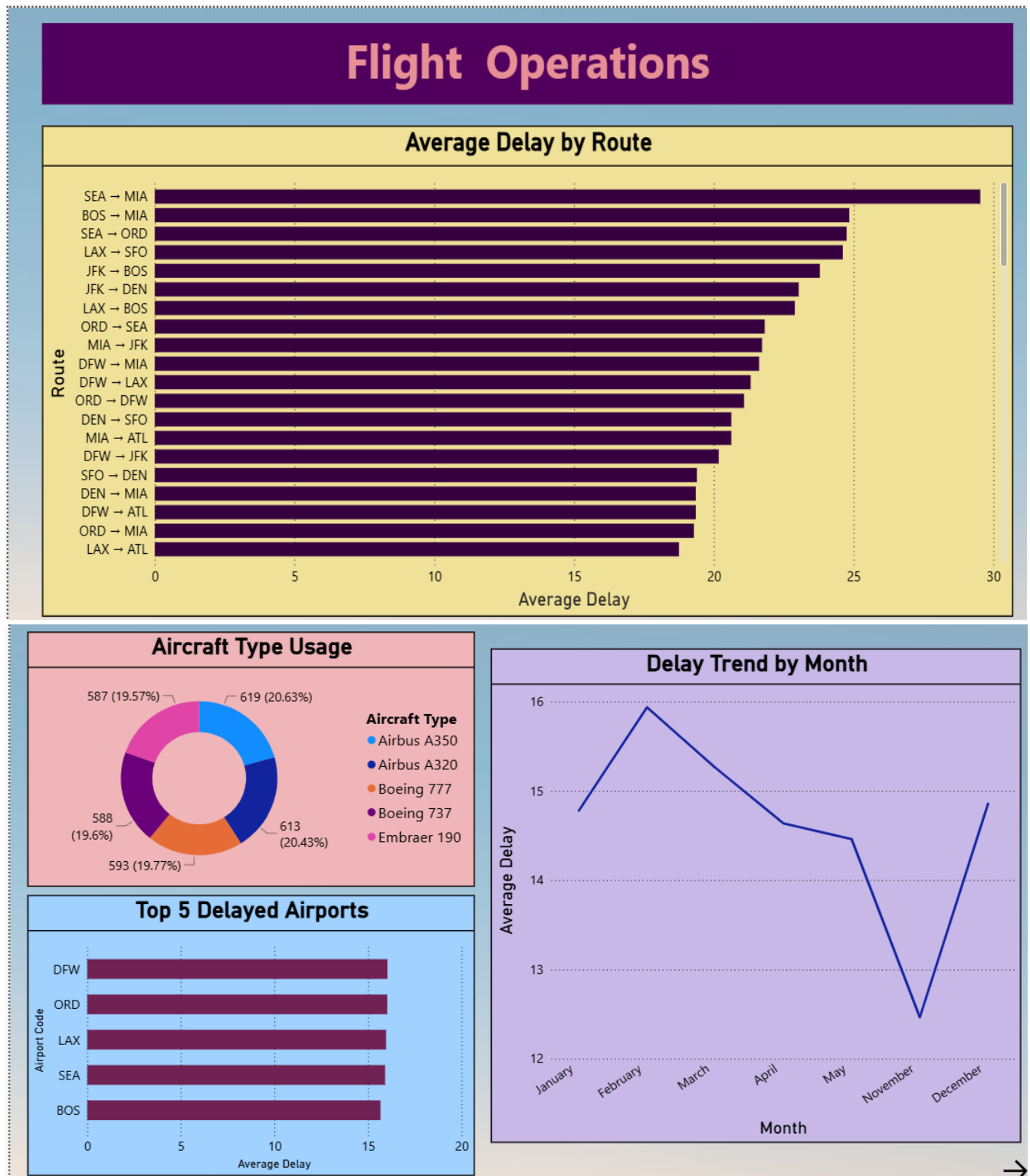
7.1 Dashboard 1: Executive Summary



Analysis:

The Executive Summary Dashboard provides a high-level overview of airline performance through key metrics such as Total Flights, Total Passengers, Total Revenue, Frequent Flyer Rate %, On Time Rate %, Average Passenger Satisfaction Score and Average Delay. It helps management monitor overall business performance and identify revenue trends.

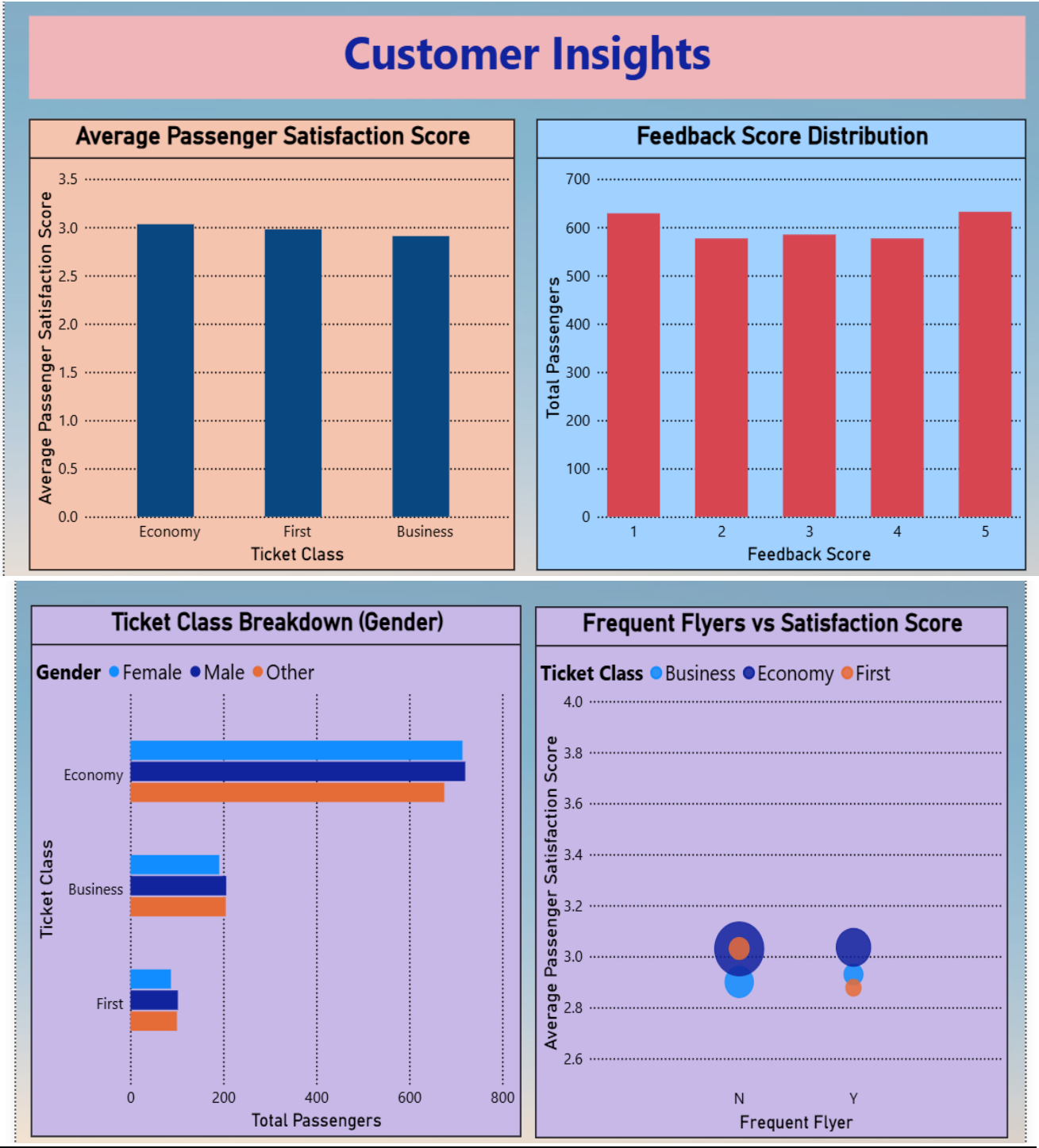
7.2 Dashboard 2: Flight Operations



Analysis:

This dashboard focuses on operational efficiency by analyzing flight delays, route performance, aircraft utilization, delay trends by month and delayed airports. It helps management identify bottlenecks and improve scheduling.

7.3 **Dashboard 3:** Customer Insights



Analysis:
The Customer Insights Dashboard evaluates passenger satisfaction, ticket class preferences, and loyalty program effectiveness. It helps improve customer experience and service quality.

8 ADVANTAGES

- Interactive dashboards.
- Faster decision-making.
- Improved operational visibility.
- Better customer satisfaction analysis.
- Revenue monitoring and forecasting support.
- User-friendly reporting interface.

9 FUTURE ENHANCEMENTS

1. Real-time flight tracking integration.
2. Predictive analytics for delay forecasting.
3. Machine learning-based passenger behaviour analysis.
4. Cloud deployment and automated refresh.
5. Mobile dashboard accessibility.

10 CONCLUSION

The SkyHigh Airlines Performance Analytics Dashboard successfully transforms airline operational and customer data into meaningful business insights. Through interactive visualizations, Cards, and DAX calculations, the solution enables management to monitor airline performance, improve customer satisfaction, and support strategic decision-making. The project demonstrates the effectiveness of Power BI as a Business Intelligence platform for airline analytics.

11 REFERENCES

- [1] Power BI Basics – Workflow.
- [2] Microsoft Power BI Documentation.

THANK YOU